

Practical 1: Build Data Warehouse and Explore WEKA

Objective: Install WEKA Workbench and explore its features such as preprocessing, classification, clustering, association rule mining, attribute selection, and visualization.

Step 1: Download WEKA

1. Open any web browser.
2. Search for “WEKA download”.
3. Open the official WEKA website.
4. Download WEKA version 3.8.6 or latest available version.
5. Save the installer file in Downloads folder.

Step 2: Install WEKA

1. Open the downloaded WEKA setup file.
2. Click **Next** on the setup wizard.
3. Accept the license agreement.
4. Choose installation location.
5. Click **Install**.
6. Wait for installation to complete.
7. Click **Finish** after successful installation.

Step 3: Launch WEKA

1. Open Start Menu.
2. Search for “WEKA”.
3. Open **WEKA GUI Chooser**.
4. Observe available options:
 - Explorer
 - Experimenter
 - KnowledgeFlow
 - Simple CLI

Step 4: Open Explorer

1. Click on **Explorer** option.
2. WEKA Explorer window will open.
3. Observe tabs available in Explorer:
 - Preprocess
 - Classify
 - Cluster
 - Associate
 - Select Attributes
 - Visualize

Step 5: Create Dataset Using Notepad

1. Open Notepad.
2. Type the following ARFF dataset:

```
@relation market
@attribute items string
@data
"bread,butter,jam"
"milk,bread"
"bread,butter"
"milk,jam"
"bread,milk,butter"
```

3. Click File → Save As.
4. Save file as **market.arff**.
5. Select "All Files" while saving.

Step 6: Load Dataset in WEKA

1. Go to Preprocess tab.
2. Click **Open File**.
3. Browse and select **market.arff**.
4. Click **Open**.
5. Observe dataset details such as attributes and instances.

Step 7: Explore Preprocess Tab

1. Observe list of attributes on left side.
2. Click on attributes to view statistics.
3. Observe histogram visualization.
4. Use filters section to preprocess data.
5. Apply filters if required.

Step 8: Explore Classification

1. Open **Classify** tab.
2. Click **Choose**.
3. Select:
bayes → NaiveBayes
4. Click **Start**.
5. Observe classifier output and accuracy results.

Step 9: Explore Clustering

1. Open **Cluster** tab.
2. Click **Choose**.
3. Select **SimpleKMeans**.
4. Click **Start**.
5. Observe clustering results.

Step 10: Explore Association Rule Mining

1. Open **Associate** tab.
2. Click **Choose**.
3. Select **Apriori** algorithm.
4. Click **Start**.
5. Observe generated association rules.

Step 11: Explore Attribute Selection

1. Open **Select Attributes** tab.
2. Choose attribute evaluator.
3. Click **Start**.
4. Observe selected important attributes.

Step 12: Explore Visualization

1. Open **Visualize** tab.
2. Select attributes for plotting.
3. Observe scatter plots and histograms.
4. Analyze visual patterns in dataset.

Step 13: Observe Output

1. Dataset loaded successfully.
2. Histogram generated.
3. Classification results displayed.
4. Clustering performed successfully.
5. Association rules generated.
6. Visualization graphs displayed.

Step 14: Conclusion

WEKA was installed successfully and its major functionalities such as preprocessing, classification, clustering, association rule mining, attribute selection, and visualization were explored successfully.